Employee Management System using Apex Console in Salesforce

Aim

To develop a console-based Employee Management System in Salesforce using Apex Programming that performs CRUD operations on Employee records.

Objective

The system manages Employee records stored in Salesforce custom object.

Operations performed:

- Create Employee
- Read Employee Records
- Update Employee
- Delete Employee

Software Requirements

- Salesforce Developer Org
- Developer Console
- Apex Programming Language

Custom Object Used

Object Name

```
Employee__c
```

Fields Used

Field Label	API Name	Data Type
Employee Name	Name	Text
Employee ID	Emp_ID__c	Number

Field Label	API Name	Data Type
Email	Email__c	Email
Birth Date	Birth_Date__c	Date
Department	Department__c	Text

Apex Class

```
public class EmployeeControllerconsole {

    // CREATE
    public void addEmployee(

        String name,
        Decimal empId,
        String email,
        Date birthDate,
        String department
    ) {

        Employee__c emp =
            new Employee__c();

        emp.Name = name;

        emp.Emp_ID__c = empId;

        emp.Email__c = email;

        emp.Birth_Date__c = birthDate;

        emp.Department__c = department;

        insert emp;

        System.debug(
            'Employee Inserted Successfully'
        );
    }

    // READ
    public void getEmployees() {

        List<Employee__c> empList = [

            SELECT Id,
                   Name,
```

```
                  Emp_ID__c,
                  Email__c,
                  Birth_Date__c,
                  Department__c

           FROM Employee__c
       ];

       System.debug(empList);
   }

   // UPDATE
   public void updateEmployee(

       Id empRecordId,
       String newName,
       String newDepartment
   ) {

       Employee__c emp = [

           SELECT Id,
                  Name,
                  Department__c

           FROM Employee__c

           WHERE Id = :empRecordId

           LIMIT 1
       ];

       emp.Name = newName;

       emp.Department__c = newDepartment;

       update emp;

       System.debug(
           'Employee Updated Successfully'
       );
   }

   // DELETE
   public void deleteEmployee(
       Id empRecordId
   ) {

       Employee__c emp = [

           SELECT Id
```

```
            FROM Employee__c

            WHERE Id = :empRecordId

            LIMIT 1
        ];

        delete emp;

        System.debug(
            'Employee Deleted Successfully'
        );
    }
}
```

Execute Anonymous Commands

CREATE COMMAND

```
EmployeeControllerconsole obj =
    new EmployeeControllerconsole();

obj.addEmployee(

    'Rahul Sharma',
    101,
    'rahul@gmail.com',
    Date.newInstance(2002,5,10),
    'IT'
);
```

READ COMMAND

```
EmployeeControllerconsole obj =
    new EmployeeControllerconsole();

obj.getEmployees();
```

UPDATE COMMAND

```
EmployeeControllerconsole obj =
    new EmployeeControllerconsole();

obj.updateEmployee(

    'a01dM00003jTK6LQAW',

    'Rahul Verma',

    'HR'
);
```

DELETE COMMAND

```
EmployeeControllerconsole obj =
    new EmployeeControllerconsole();

obj.deleteEmployee(
    'a01dM00003jTK6LQAW'
);
```

Steps to Execute in Developer Console

Step 1

Open Salesforce Developer Console.

```
Gear Icon
→ Developer Console
```

Step 2

Open Execute Anonymous Window.

```
Debug
→ Open Execute Anonymous Window
```

Shortcut:

```
Ctrl + E
```

Step 3

Paste required command.

- Create
- Read
- Update
- Delete

Step 4

Click:

```
Execute
```

Debugging Steps

Step 1

Open:

```
Logs
```

Step 2

Open latest debug log.

Step 3

Search:

```
USER_DEBUG
```

Sample Outputs

CREATE OUTPUT

```
Employee Inserted Successfully
```

UPDATE OUTPUT

```
Employee Updated Successfully
```

DELETE OUTPUT

```
Employee Deleted Successfully
```

READ OUTPUT

```
(Employee__c:{
Name=Rahul Sharma,
Department__c=IT
})
```

Concepts Used

Concept	Purpose
Apex	Backend Programming
SOQL	Fetch Records
DML	Insert, Update, Delete
Execute Anonymous	Execute Apex Code
System.debug()	Display Output

Advantages

- Simple console-based CRUD system
- Easy record management
- Fast execution using Apex
- Useful for learning Salesforce database operations

Result

Thus, the console-based Employee Management System was successfully developed in Salesforce using Apex Programming and CRUD operations were performed successfully.